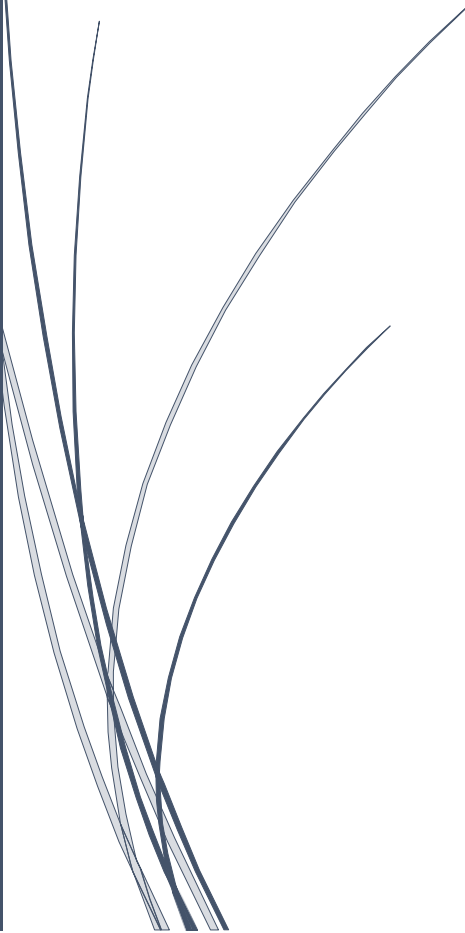


A dark blue vertical bar on the left side of the page. A blue arrow points to the right from the bar, containing the text 'Mr Michael Weiss'.

Mr Michael Weiss

Network Service & Virtualization

Proof of Concept Linux virtual
network Project

Several thin, dark blue wavy lines that originate from the bottom left and curve upwards and to the right.

NOME: Marco dos Santos
STUDENT NUMBER: 2020333

Summary

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1- Obtain Linux updated and ping to the servers

Ubuntuserver update

```
msantos@ubuntuserver:~$ sudo apt install update
[sudo] password for msantos:
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package update
msantos@ubuntuserver:~$ sudo apt update
Hit:1 http://ie.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://ie.archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB]
Get:3 http://ie.archive.ubuntu.com/ubuntu focal-backports InRelease [101 kB]
Get:4 http://ie.archive.ubuntu.com/ubuntu focal-security InRelease [109 kB]
Get:5 http://ie.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [951 kB]
Get:6 http://ie.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [765 kB]
Fetched 2,040 kB in 8s (262 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
88 packages can be upgraded. Run 'apt list --upgradable' to see them.
msantos@ubuntuserver:~$
```

Ubuntuclient update

```
msantos@ubuntuclient:~$ sudo apt install update
[sudo] password for msantos:
Sorry, try again.
[sudo] password for msantos:
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package update
msantos@ubuntuclient:~$ sudo apt update
Hit:1 http://ie.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://ie.archive.ubuntu.com/ubuntu focal-updates InRelease
Get:3 http://ie.archive.ubuntu.com/ubuntu focal-backports InRelease [101 kB]
Get:4 http://ie.archive.ubuntu.com/ubuntu focal-security InRelease [109 kB]
Get:5 http://ie.archive.ubuntu.com/ubuntu focal/main Translation-en_GB [485 kB]
Get:6 http://ie.archive.ubuntu.com/ubuntu focal/restricted Translation-en_GB [3,860 B]
Get:7 http://ie.archive.ubuntu.com/ubuntu focal/universe Translation-en_GB [319 kB]
Get:8 http://ie.archive.ubuntu.com/ubuntu focal/multiverse Translation-en_GB [105 kB]
Fetched 1,123 kB in 7s (155 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
59 packages can be upgraded. Run 'apt list --upgradable' to see them.
msantos@ubuntuclient:~$ _
```

Ubuntu server upgrade

```
E: Unable to locate package upgrade
msantos@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-5.4.0-65 linux-headers-5.4.0-65-generic linux-image-5.4.0-65-generic
  linux-modules-5.4.0-65-generic linux-modules-extra-5.4.0-65-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libbevd2 libimobiledevice6 libplist3 libupower-glib3 libusbmuxd6 linux-headers-5.4.0-72
  linux-headers-5.4.0-72-generic linux-image-5.4.0-72-generic linux-modules-5.4.0-72-generic
  linux-modules-extra-5.4.0-72-generic upower usbmuxd
The following packages will be upgraded:
  alsa-ucm-conf apport apt apt-utils bind9-dnsutils bind9-host bind9-libs ca-certificates
  cloud-init curl dirmngr distro-info-data friendly-recovery git git-man gnupg gnupg-l10n
  gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv grub-common grub-pc
  grub-pc-bin grub2-common initramfs-tools initramfs-tools-bin initramfs-tools-core
  isc-dhcp-client isc-dhcp-common landscape-common libapt-pkg6.0 libasound2 libasound2-data
  libcurl3-gnutls libcurl4 libglib2.0-0 libglib2.0-bin libglib2.0-data libhogweed5 libldap-2.4-2
  libldap-common libnettle7 libnss-systemd libpam-systemd libpci3 libprocps8 libpython3.8
  libpython3.8-minimal libpython3.8-stdlib libseccomp2 libssl1.1 libsystemd0 libudev1 libxmlb1
  libzstd1 linux-firmware linux-generic linux-headers-generic linux-image-generic open-iscsi
  open-vm-tools openssh-client openssh-server openssh-sftp-server openssl pciutils pollinate
  procps python3-apport python3-distupgrade python3-problem-report python3-software-properties
  python3-twisted python3-twisted-bin python3-update-manager python3.8 python3.8-minimal screen snapd
  software-properties-common sosreport systemd systemd-sysv systemd-timesyncd thermald tmux
  ubuntu-keyring ubuntu-release-upgrader-core udev update-manager-core update-notifier-common
96 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 248 MB of archives.
After this operation, 382 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

Ubuntu client upgrade

```
E: Unable to locate package upgrade
msantos@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-5.4.0-65 linux-headers-5.4.0-65-generic linux-image-5.4.0-65-generic
  linux-modules-5.4.0-65-generic linux-modules-extra-5.4.0-65-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libbevd2 libimobiledevice6 libplist3 libupower-glib3 libusbmuxd6 linux-headers-5.4.0-72
  linux-headers-5.4.0-72-generic linux-image-5.4.0-72-generic linux-modules-5.4.0-72-generic
  linux-modules-extra-5.4.0-72-generic upower usbmuxd
The following packages will be upgraded:
  alsa-ucm-conf apport apt apt-utils bind9-dnsutils bind9-host bind9-libs ca-certificates
  cloud-init curl dirmngr distro-info-data friendly-recovery git git-man gnupg gnupg-l10n
  gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv grub-common grub-pc
  grub-pc-bin grub2-common initramfs-tools initramfs-tools-bin initramfs-tools-core
  isc-dhcp-client isc-dhcp-common landscape-common libapt-pkg6.0 libasound2 libasound2-data
  libcurl3-gnutls libcurl4 libglib2.0-0 libglib2.0-bin libglib2.0-data libhogweed5 libldap-2.4-2
  libldap-common libnettle7 libnss-systemd libpam-systemd libpci3 libprocps8 libpython3.8
  libpython3.8-minimal libpython3.8-stdlib libseccomp2 libssl1.1 libsystemd0 libudev1 libxmlb1
  libzstd1 linux-firmware linux-generic linux-headers-generic linux-image-generic open-iscsi
  open-vm-tools openssh-client openssl pciutils pollinate procps python3-apport
  python3-distupgrade python3-problem-report python3-software-properties python3-twisted
  python3-twisted-bin python3-update-manager python3.8 python3.8-minimal screen snapd
  software-properties-common sosreport systemd systemd-sysv systemd-timesyncd thermald tmux
  ubuntu-keyring ubuntu-release-upgrader-core udev update-manager-core update-notifier-common
94 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 247 MB of archives.
After this operation, 382 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

Ifconfig ubuntuserver

```
msantos@ubuntuserver:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fe75:a118 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:75:a1:18 txqueuelen 1000 (Ethernet)
    RX packets 87 bytes 19030 (19.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 32 bytes 4246 (4.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.3.15 netmask 255.255.255.0 broadcast 10.0.3.255
    inet6 fe80::a00:27ff:fe33:4657 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:33:46:57 txqueuelen 1000 (Ethernet)
    RX packets 41131 bytes 60876286 (60.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3238 bytes 210263 (210.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 126 bytes 10582 (10.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 126 bytes 10582 (10.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Ifconfig ubuntuclient

```
msantos@ubuntuclient:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fe1d:dd82 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:1d:dd:82 txqueuelen 1000 (Ethernet)
    RX packets 88 bytes 19620 (19.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34 bytes 4654 (4.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.3.15 netmask 255.255.255.0 broadcast 10.0.3.255
    inet6 fe80::a00:27ff:fe78:31e0 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:78:31:e0 txqueuelen 1000 (Ethernet)
    RX packets 43255 bytes 60077410 (60.0 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8436 bytes 521117 (521.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 126 bytes 10582 (10.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 126 bytes 10582 (10.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Server pinging client

```
msantos@ubuntuserver:~$ ping 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data:
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=0.899 ms
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=0.888 ms
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=0.929 ms
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=0.910 ms
64 bytes from 192.168.56.104: icmp_seq=5 ttl=64 time=0.904 ms
64 bytes from 192.168.56.104: icmp_seq=6 ttl=64 time=0.916 ms
64 bytes from 192.168.56.104: icmp_seq=7 ttl=64 time=0.610 ms
64 bytes from 192.168.56.104: icmp_seq=8 ttl=64 time=0.810 ms
^C
--- 192.168.56.104 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7011ms
rtt min/avg/max/mdev = 0.610/0.858/0.929/0.099 ms
msantos@ubuntuserver:~$ _
```

Install apache and display apache page.

Installing apache in the server

```
Processing triggers for libc-bin (2.31-0ubuntu9.5) ...
msantos@ubuntuserver:~$ sudo apt install apache2 -y_
```

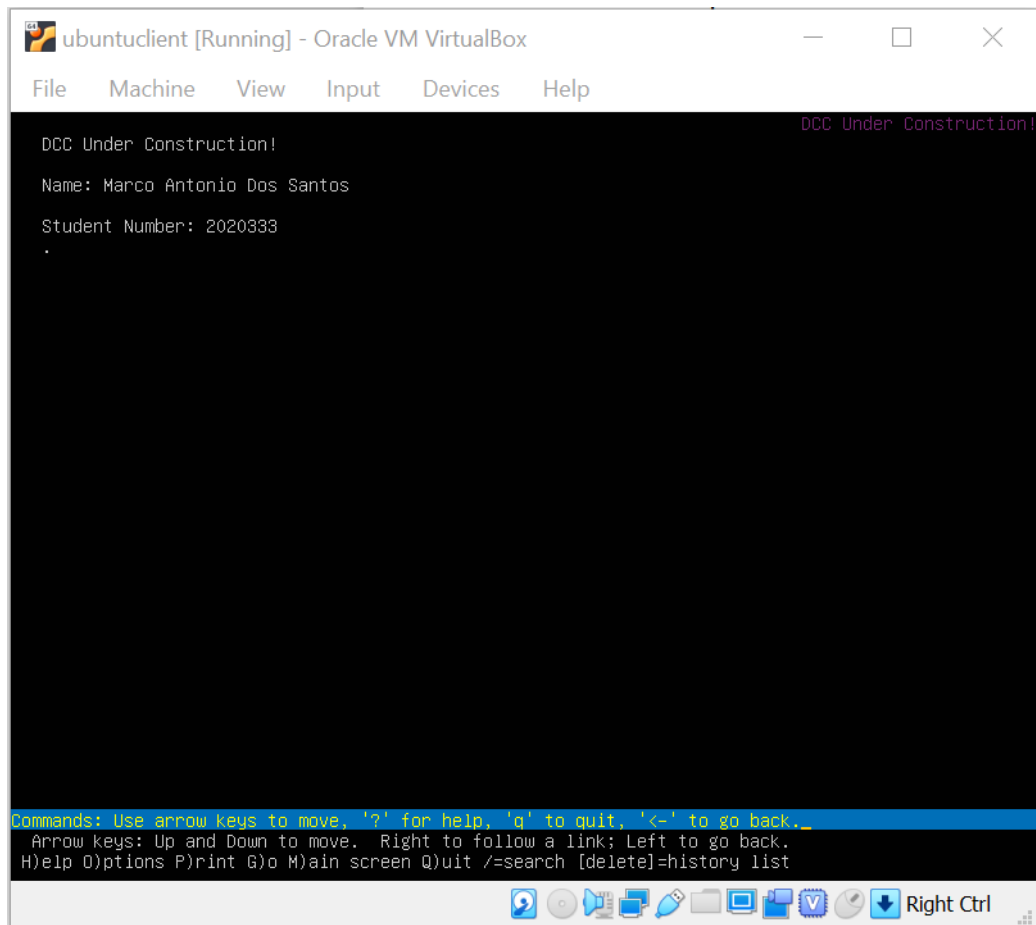
Installing lynx in the client

```
Processing triggers for mime-support (3.04ubuntu1) ...
msantos@ubuntucient:~$ sudo apt install lynx -y_
```

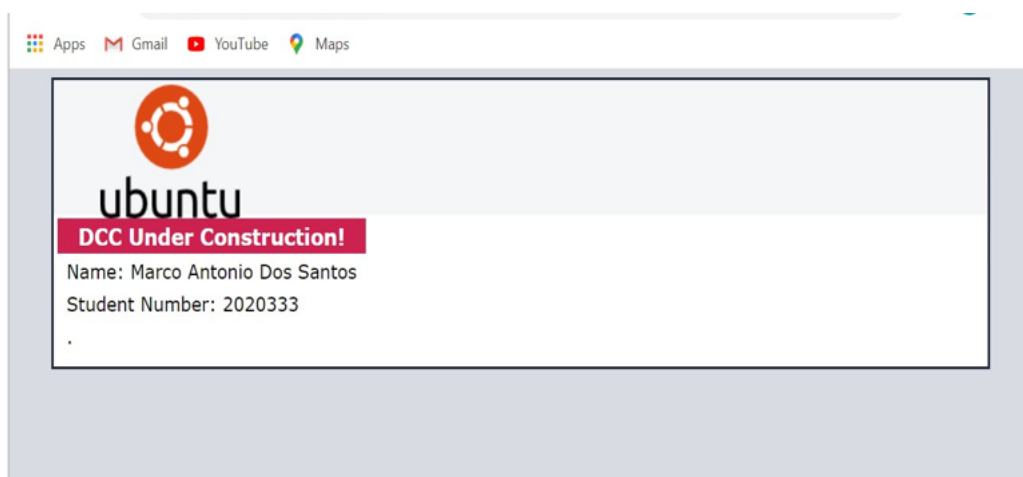
Client displaying apache page

```
msantos@ubuntucient:~$ lynx www.apache.com
```

Displayed test to apache home page!



Host operating system to access to the Apache home page.



Host machine pinging server machine by command prompt.

```
C:\WINDOWS\system32>ping 192.168.56.103

Pinging 192.168.56.103 with 32 bytes of data:
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.56.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\WINDOWS\system32>
```

WIRESHARK

ICMP Traffic - host machine pinging server machine

VirtualBox Host-Only Network

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.56.1	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (reply in 2)
2	0.000276	192.168.56.103	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=17/4352, ttl=64 (request in 1)
3	1.002567	192.168.56.1	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=18/4608, ttl=128 (reply in 4)
4	1.003028	192.168.56.103	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=18/4608, ttl=64 (request in 3)
5	2.008815	192.168.56.1	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=19/4864, ttl=128 (reply in 6)
6	2.009178	192.168.56.103	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=19/4864, ttl=64 (request in 5)
7	3.013128	192.168.56.1	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=20/5120, ttl=128 (reply in 8)
8	3.013389	192.168.56.103	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=20/5120, ttl=64 (request in 7)

http traffic after displaying apache webpage in google chrome browser

*VirtualBox Host-Only Network

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
7	0.004563	192.168.56.1	192.168.56.103	HTTP	591	GET / HTTP/1.1
11	0.005819	192.168.56.103	192.168.56.1	HTTP	611	HTTP/1.1 200 OK (text/html)

> Frame 7: 591 bytes on wire (4728 bits), 591 bytes captured (4728 bits) on interface \Device\NPF_{4CB9034F-906C-4C83-9CDF-E75E626A778B}, id 0
> Ethernet II, Src: 0a:00:27:00:00:07 (0a:00:27:00:00:07), Dst: PcsCompu_75:a1:18 (08:00:27:75:a1:18)
> Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.103
> Transmission Control Protocol, Src Port: 51141, Dst Port: 80, Seq: 1, Ack: 1, Len: 537
> Hypertext Transfer Protocol

3 way handshake

*VirtualBox Host-Only Network

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.56.1	192.168.56.103	TCP	66	66 51141 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
2	0.000355	192.168.56.103	192.168.56.1	TCP	66	80 → 51141 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
3	0.000459	192.168.56.1	192.168.56.103	TCP	66	66 51142 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
4	0.000532	192.168.56.1	192.168.56.103	TCP	54	54 51141 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
5	0.000630	192.168.56.103	192.168.56.1	TCP	66	80 → 51142 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
6	0.000749	192.168.56.1	192.168.56.103	TCP	54	54 51142 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
7	0.004563	192.168.56.1	192.168.56.103	HTTP	591	GET / HTTP/1.1
8	0.004756	192.168.56.103	192.168.56.1	TCP	60	80 → 51141 [ACK] Seq=1 Ack=538 Win=64128 Len=0
9	0.005759	192.168.56.103	192.168.56.1	TCP	1514	80 → 51141 [ACK] Seq=1 Ack=538 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
10	0.005811	192.168.56.103	192.168.56.1	TCP	1514	80 → 51141 [ACK] Seq=1461 Ack=538 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
11	0.005819	192.168.56.103	192.168.56.1	HTTP	611	HTTP/1.1 200 OK (text/html)
12	0.006614	192.168.56.1	192.168.56.103	TCP	54	54 51141 → 80 [ACK] Seq=538 Ack=3478 Win=262656 Len=0

> Frame 7: 591 bytes on wire (4728 bits), 591 bytes captured (4728 bits) on interface \Device\NPF_{4CB9034F-906C-4C83-9CDF-E75E626A778B}, id 0
> Ethernet II, Src: 0a:00:27:00:00:07 (0a:00:27:00:00:07), Dst: PcsCompu_75:a1:18 (08:00:27:75:a1:18)
> Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.103
> Transmission Control Protocol, Src Port: 51141, Dst Port: 80, Seq: 1, Ack: 1, Len: 537
> Hypertext Transfer Protocol

2-SSH logging and encrypted traffic

SSH installed

```
msantos@ubuntuuserver:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2022-05-16 17:00:20 UTC; 14s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 17828 (sshd)
    Tasks: 1 (limit: 1066)
   Memory: 1.3M
   CGroup: /system.slice/ssh.service
           └─17828 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

May 16 17:00:20 ubuntuuserver systemd[1]: Starting OpenBSD Secure Shell server...
May 16 17:00:20 ubuntuuserver sshd[17828]: Server listening on 0.0.0.0 port 22.
May 16 17:00:20 ubuntuuserver sshd[17828]: Server listening on :: port 22.
May 16 17:00:20 ubuntuuserver systemd[1]: Started OpenBSD Secure Shell server.
msantos@ubuntuuserver:~$
```

Accessing SSH

```
msantos@ubuntuuserver: ~
login as: msantos
msantos@192.168.56.103's password:
Welcome to Ubuntu 20.04.4 LTS (GNU/Linux 5.4.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Mon 16 May 17:58:25 UTC 2022

System load:  0.0               Processes:    108
Usage of /:   45.1% of 8.90GB   Users logged in: 1
Memory usage: 24%              IPv4 address for enp0s3: 192.168.56.103
Swap usage:   0%               IPv4 address for enp0s8: 10.0.3.15

0 updates can be applied immediately.
```

SSH traffic in Wireshark

VirtualBox Host-Only Network

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ssh

No.	Time	Source	Destination	Protocol	Length	Info
25	46.532684	192.168.56.1	192.168.56.103	SSHv2	82	Client: Protocol (SSH-2.0-PuTTY_Release_0.76)
27	46.536448	192.168.56.103	192.168.56.1	SSHv2	95	Server: Protocol (SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.5)
28	46.539698	192.168.56.1	192.168.56.103	SSHv2	1310	Client: Key Exchange Init
29	46.539873	192.168.56.103	192.168.56.1	SSHv2	1110	Server: Key Exchange Init
30	46.542045	192.168.56.1	192.168.56.103	SSHv2	102	Client: Elliptic Curve Diffie-Hellman Key Exchange Init
32	46.546396	192.168.56.103	192.168.56.1	SSHv2	518	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=256)
33	46.554671	192.168.56.1	192.168.56.103	SSHv2	134	Client: New Keys, Encrypted packet (len=64)
35	46.554873	192.168.56.103	192.168.56.1	SSHv2	118	Server: Encrypted packet (len=64)
39	52.191460	192.168.56.1	192.168.56.103	SSHv2	134	Client: Encrypted packet (len=80)
40	52.199655	192.168.56.103	192.168.56.1	SSHv2	134	Server: Encrypted packet (len=80)
51	62.631223	192.168.56.1	192.168.56.103	SSHv2	326	Client: Encrypted packet (len=272)
52	62.639401	192.168.56.103	192.168.56.1	SSHv2	102	Server: Encrypted packet (len=48)

> Frame 25: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Device\NPF_{4CB9034F-906C-4C83-9CDF-E75E626A7788}, id 0

> Ethernet II, Src: 0a:00:27:00:00:07 (0a:00:27:00:00:07), Dst: PcsCompu_75:a1:18 (08:00:27:75:a1:18)

> Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.103

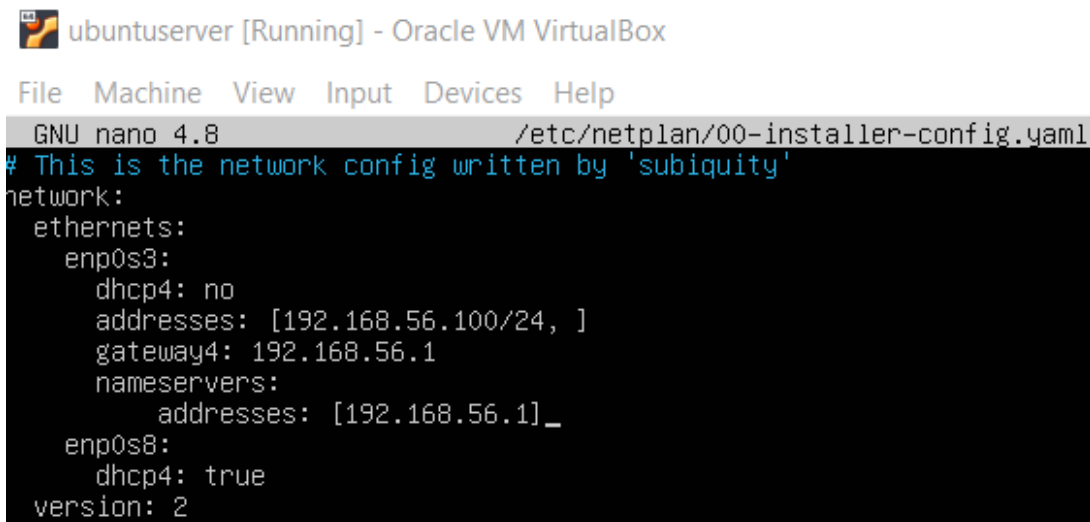
> Transmission Control Protocol, Src Port: 56749, Dst Port: 22, Seq: 1, Ack: 1, Len: 28

> SSH Protocol

3-IP address and Hostname management

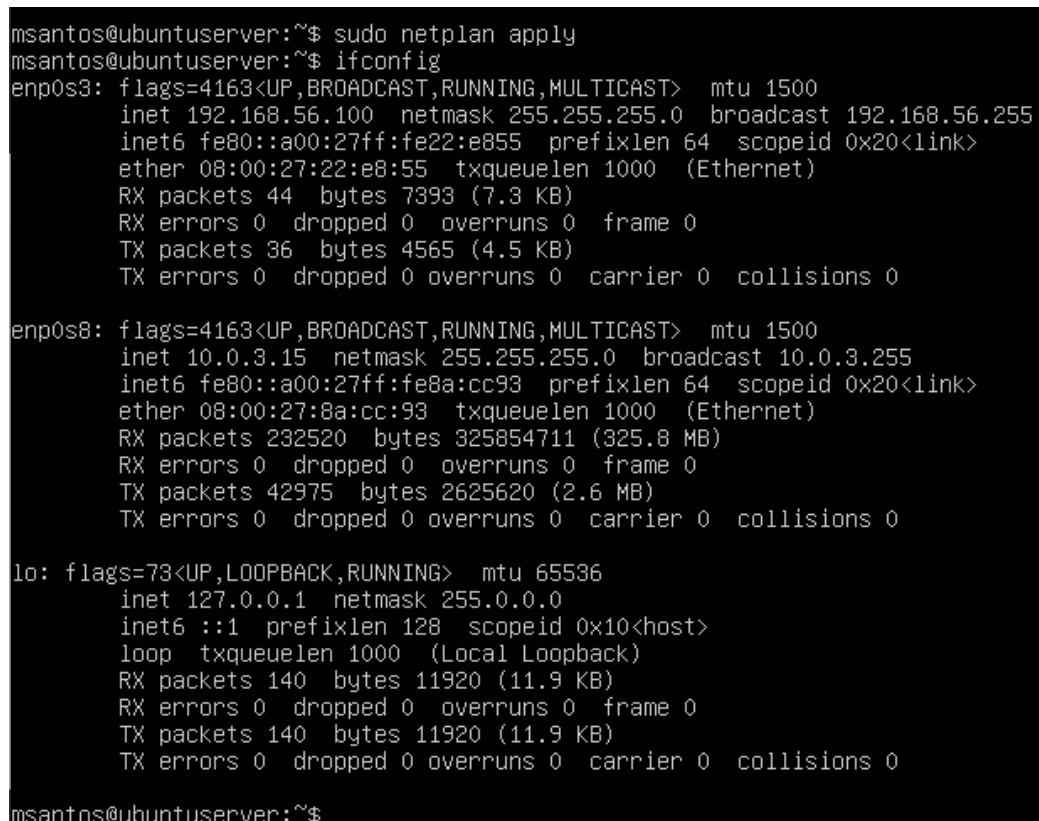
According to Daniel(2022):” Configuring a static IP can be difficult in Linux because it’s different based on the distro and version you’re using.”

Configuring server using Netplan.



```
ubuntuserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 4.8 /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: no
      addresses: [192.168.56.100/24, ]
      gateway4: 192.168.56.1
      nameservers:
        addresses: [192.168.56.1]_
    enp0s8:
      dhcp4: true
  version: 2
```

Sudo Netplan apply: new Ip address set



```
msantos@ubuntuserver:~$ sudo netplan apply
msantos@ubuntuserver:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.100 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fe22:e855 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:22:e8:55 txqueuelen 1000 (Ethernet)
    RX packets 44 bytes 7393 (7.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 36 bytes 4565 (4.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.3.15 netmask 255.255.255.0 broadcast 10.0.3.255
    inet6 fe80::a00:27ff:fe8a:cc93 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8a:cc:93 txqueuelen 1000 (Ethernet)
    RX packets 232520 bytes 325854711 (325.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 42975 bytes 2625620 (2.6 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 140 bytes 11920 (11.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 140 bytes 11920 (11.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

msantos@ubuntuserver:~$
```

Changing ip address of Ubuntuclient using Netplan.

```
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: no
      addresses: [192.168.56.125/24, ]
      gateway4: 192.168.56.1
      nameservers:
        addresses: [192.168.56.1]
    enp0s8:
      dhcp4: true
  version: 2
```

Change was successful

```
msantos@ubuntuclient:~$ sudo netplan apply
msantos@ubuntuclient:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.125 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fe1d:dd82 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:1d:dd:82 txqueuelen 1000 (Ethernet)
    RX packets 890 bytes 136014 (136.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 201 bytes 32891 (32.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.3.15 netmask 255.255.255.0 broadcast 10.0.3.255
    inet6 fe80::a00:27ff:fe78:31e0 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:78:31:e0 txqueuelen 1000 (Ethernet)
    RX packets 50780 bytes 70580785 (70.5 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10044 bytes 631708 (631.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 212 bytes 18388 (18.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 212 bytes 18388 (18.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

msantos@ubuntuclient:~$
```

Rename Hostname(s)

Ubuntuservers rename

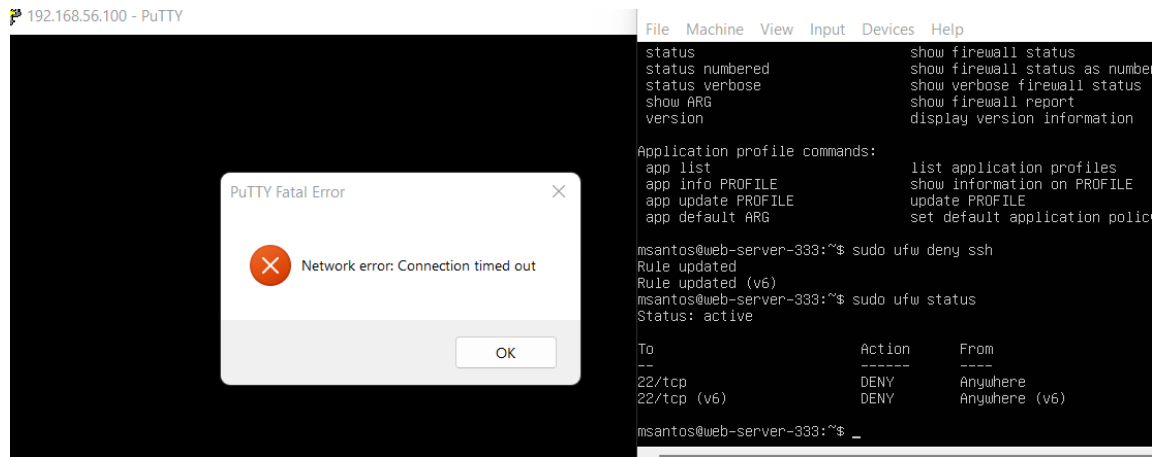
```
root@ubuntuserver:/home/msantos# cat /etc/hostname
ubuntuserver
root@ubuntuserver:/home/msantos# hostnamectl set-hostname web-server-333
root@ubuntuserver:/home/msantos# hostname
web-server-333
root@ubuntuserver:/home/msantos# hostnamectl
  Static hostname: web-server-333
            Icon name: computer-vm
            Chassis: vm
            Machine ID: 40621680576c401a8827ef4d7fbeab67
            Boot ID: f0ca8fbbb88f4b2193ciabc871dfe6c0
    Virtualization: oracle
  Operating System: Ubuntu 20.04.2 LTS
            Kernel: Linux 5.4.0-65-generic
    Architecture: x86-64
root@ubuntuserver:/home/msantos# hostname
web-server-333
```

Ubuntuclient rename

```
root@ubuntuclient:/home/msantos# cat /etc/hostname
ubuntuclient
root@ubuntuclient:/home/msantos# hostnamectl set-hostname web-client-333
root@ubuntuclient:/home/msantos# hostname
web-client-333
root@ubuntuclient:/home/msantos# hostnamectl
  Static hostname: web-client-333
            Icon name: computer-vm
            Chassis: vm
            Machine ID: 9a0687241d3e4aecadb9047e9e74d4cc
            Boot ID: b87e49cf7ae84c60ab1eaa98ae091950
    Virtualization: oracle
  Operating System: Ubuntu 20.04.2 LTS
            Kernel: Linux 5.4.0-72-generic
    Architecture: x86-64
root@ubuntuclient:/home/msantos# hostname
web-client-333
```

4-Firewall

Permission denied. Firewall is active but denying SSH.



Firewall is active. SSH access granted



The same process was repeated in UbuntuClient.

Reference:

Miessler,D.,2022.How to Set a Static IP Address in Linux .[Online] Available at:
<https://danielmiessler.com/study/manually-set-ip-linux/> ,[Accessed 19 May
2022].